



TF

RECOVERY

# Recovery & CNS Restoration

Sleep + HRV Guide

42 Pages

■10,000 Value

Watermarked

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India's Premium Fitness Knowledge Platform

Chapter 1: CNS Stress Model — Sympathetic vs Parasympathetic

The Central Nervous System (CNS) operates through two branches: the Sympathetic (fight-or-flight) and Parasympathetic (rest-and-digest). Recovery and anabolism primarily occur in the parasympathetic state. Chronic sympathetic dominance (overtraining, life stress, poor sleep) suppresses muscle growth and hormonal function.

| Sympathetic Activation (During/After Training) | Parasympathetic Recovery State           |
|------------------------------------------------|------------------------------------------|
| • Cortisol + Adrenaline elevated               | • Cortisol returns to baseline           |
| • Heart rate + blood pressure elevated         | • Heart rate variability (HRV) increases |
| • Blood flow to muscles increased              | • Growth hormone peaks during deep sleep |
| • Digestion slowed                             | • Protein synthesis maximised            |
| • Anabolism paused                             | • Muscle repair and growth occurs        |
| • Performance output maximum                   | • Glycogen replenishment accelerated     |
| • Duration after training: 2–4 hours           | • Requires: sleep, nutrition, low stress |

Chapter 2: Sleep Stages & Recovery Diagram

| Sleep Stage          | Duration/Night         | Recovery Benefit                       | Disrupted By         | Optimise By                |
|----------------------|------------------------|----------------------------------------|----------------------|----------------------------|
| Stage 1 (Light)      | 5–10 min/cycle         | Transition, minimal                    | Noise, light         | Dark room, earplugs        |
| Stage 2 (Light NREM) | 20 min/cycle           | Consolidation, temp drop               | Stress, caffeine     | Cool room 18–20°C          |
| Stage 3 (Deep NREM)  | 20–40 min early night  | GH release, tissue repair, immune      | Alcohol, late eating | Magnesium, ZMA             |
| REM Sleep            | 90–120 min/night total | Cognitive recovery, hormone regulation | Stimulants, alcohol  | Consistent bed time ±30min |

Chapter 3: Recovery Modalities — Evidence Rating

| Modality                 | Mechanism                                                    | Evidence Level  | Protocol                                | Best Timing                             |
|--------------------------|--------------------------------------------------------------|-----------------|-----------------------------------------|-----------------------------------------|
| Sleep (7–9h)             | GH release, T production, protein synthesis, CNS restoration | ★★★★★ (Highest) | 7–9h consistent schedule                | Every night non-negotiable              |
| Nutrition (protein+carb) | mTOR activation, glycogen replenishment, anabolism           | ★★★★★           | Protein 40g + carbs 60g within 60min    | Within 60 min post-workout              |
| Active Recovery          | Blood flow, lactate clearance, ROM maintenance               | ★★★★            | 20–30 min walk/swim/yoga at 40–50% HR   | Day after heavy training                |
| Cold Water Immersion     | ↓ inflammation, ↓ DOMS, vasoconstriction → vasodilation      | ★★★★            | Cold (10–15°C) 10–15 min, 3–4x/week     | Post-workout, not post-strength session |
| Sauna (Finnish dry)      | Heat shock proteins, cardiovascular, GH release              | ★★★★            | 80–90°C, 15–20 min, 3–4x/week           | 3–4h post-workout or rest days          |
| Foam Rolling/Massage     | Myofascial release, blood flow, DOMS reduction               | ★★★             | 10–15 min post-workout on target muscle | Post-session or rest days               |
| Compression garments     | ↓ oedema, blood pooling reduction, faster DOMS clearance     | ★★★             | Wear post-training 2–4h                 | Post-workout period                     |
| Contrast therapy         | Alternating cold/heat vasodilation-vasoconstriction          | ★★★             | Cold 1 min → Hot 3 min, 3–4 cycles      | Post-competition or hard week           |

#### Chapter 4: HRV (Heart Rate Variability) — Your Recovery Metric

HRV measures the variation in time between successive heartbeats. Higher HRV = better parasympathetic recovery. Elite athletes track HRV daily to guide training load.

| HRV Status         | Score (RMSSD)          | Training Recommendation                     | Action                                           |
|--------------------|------------------------|---------------------------------------------|--------------------------------------------------|
| Excellent Recovery | >70ms (individualised) | Train hard — CNS is ready                   | Full intensity session planned                   |
| Good Recovery      | 60–70ms                | Normal training — moderate intensity        | Stick to program as planned                      |
| Moderate Recovery  | 50–60ms                | Reduce volume by 20%, avoid maximal effort  | Drop 1–2 sets, skip HIIT                         |
| Poor Recovery      | 40–50ms                | Active recovery only — walk, yoga, mobility | No lifting today                                 |
| Very Poor Recovery | <40ms                  | Complete rest — something is wrong          | Sleep, nutrition audit, see doctor if persistent |

## Chapter 5: Supplement Stack for Recovery

| Supplement           | Dosage & Timing                        | Evidence | Primary Benefit                                                  |
|----------------------|----------------------------------------|----------|------------------------------------------------------------------|
| Magnesium Glycinate  | 400–600mg before bed                   | ★★★★★    | Deep sleep (Stage 3 NREM), muscle relaxation, cortisol reduction |
| ZMA (Zinc+Mg+B6)     | ZMA formula before bed (empty stomach) | ★★★★★    | T maintenance, sleep quality, recovery enhancement               |
| Ashwagandha KSM-66   | 600mg with dinner                      | ★★★★★    | Cortisol ↓ 28%, stress response ↓, sleep quality ↑               |
| Melatonin            | 0.5–2mg (low dose) 30min before bed    | ★★★★★    | Sleep onset, circadian rhythm regulation                         |
| L-Theanine           | 200–400mg before bed or post-training  | ★★★★★    | Alpha brain wave increase, calm focus, stress reduction          |
| Tart Cherry Extract  | 480mg twice daily                      | ★★★★★    | ↓ DOMS 20–30%, anti-inflammatory, sleep quality                  |
| Omega-3 (EPA+DHA)    | 2–4g EPA+DHA daily with food           | ★★★★★    | Inflammation resolution, joint recovery, brain recovery          |
| Creatine Monohydrate | 5g/day anytime                         | ★★★★★    | Cellular energy, strength maintenance, reduces fatigue           |

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